Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 4.0, **Version Date** 5/22/08

Meter Verification (16, Ultra, Plus)

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	40 mins	Not Applicable

2 Overview

Objective: Verify that the internal mA meter are functional. If the meter is not functional, the inverter cannot be used in the system.

- Connect meter to test points and a load resistor for the tube current
- Perform x-ray exposures
- Take measurements

Last Revised:	22 May 2008

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
HP60 SV IDIOSYNCRATIC Tools	1	-	2404264	-

3.2 Safety



⚠CAUTION
X-ray Exposure.
The procedure described in this document requires performing x-ray exposures, unlike the Meter Verification procedure for Pro16, which has a function to verify the internal mA and kV meters without performing x-ray exposures.

4 Procedure

1. Perform warm-up scans, if the x-ray tube is cold.
2. Remove gantry right side cover if it's not already removed.
3. Turn off all three service switches on the Service Switch Panel (AXIAL DRIVE ENABLE, HVDC ENABLE, 120VAC).
4. Rotate the gantry such that the power unit is in the 2 O'clock position.
5. Remove the Inverter cover. If there are two covers on the inverter, remove the larger of the two, nearest the cradle.
6. Lock the gantry in position, using the rotational lock. Ensure that gantry rotation is locked by attempting to rotate the gantry by hand.
7. Attach a protective ESD wrist strap.

NOTICE

Potential for Equipment Damage. Failure to use protective ESD equipment may result in destroyed/damages equipment.

8. Connect a 'JEDI Shunt Measurement Tool' and a multi-meter (see Illustration 1, Illustration 2, and Illustration 3)
    a. Disconnect the flat cable from J1 connector on the kV Control board.
    b. Connect J1 of the JEDI Shunt Measurement Tool to the J1 connector on the kV Control board.
    c. Connect the flat cable to J3 of the JEDI Shunt Measurement Tool.
    d. Connect a digital multi-meter (DMM/DVM) to J4 of the JEDI Shunt Measurement Tool.

Illustration 1: JEDI Shunt Measurement Tool

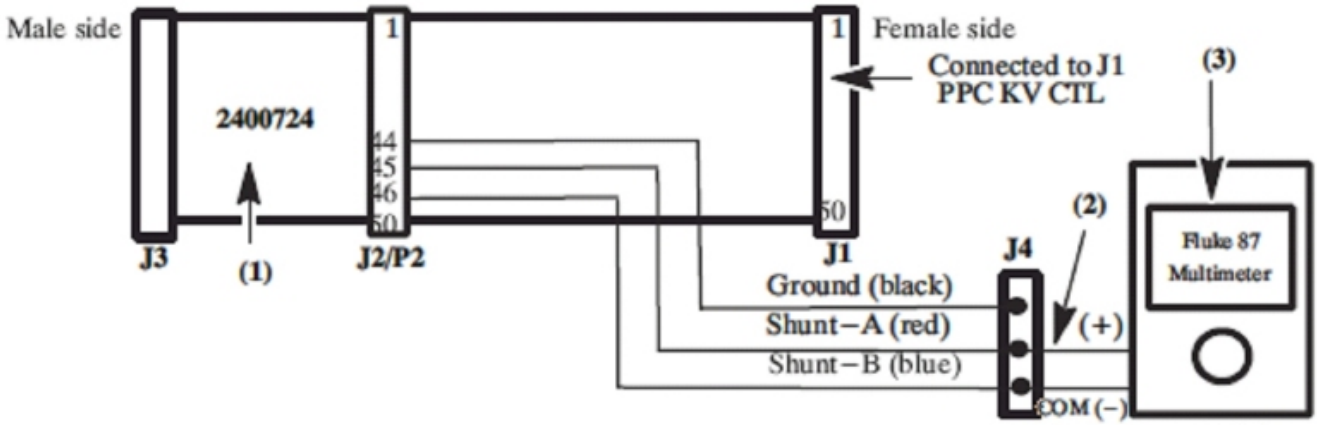


Illustration 2: Connection of the JEDI Shunt Measurement Tool

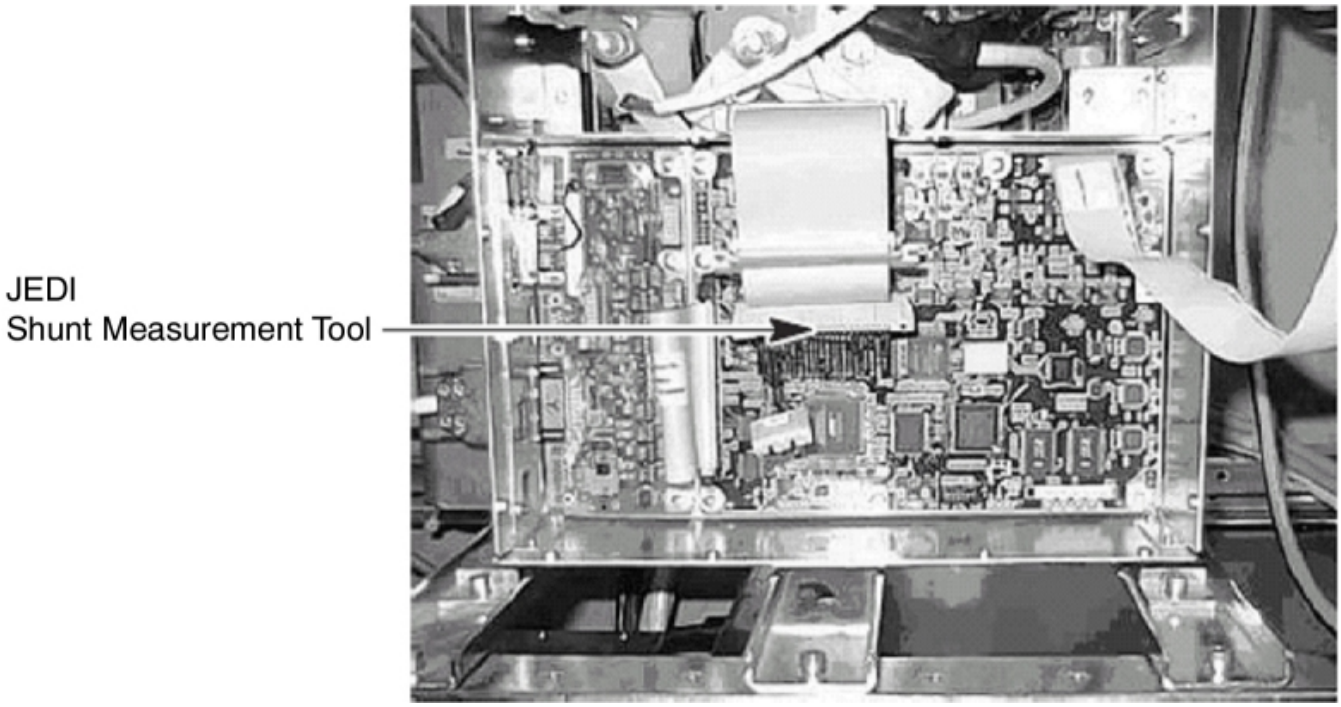
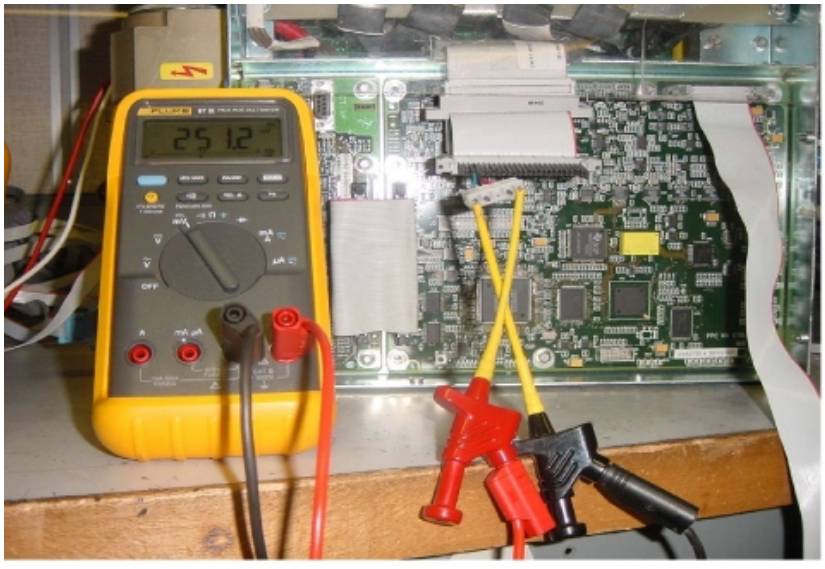


Illustration 3: Connection of the Multi-meter





9. Measure, and fine the value of shunt resistor :
    a. Set the multi-meter to the Ohmmeter function.
    b. Connect the multi-meter between Ground (black pin) and Shunt A (red pin) on the external tool.
    c. Record the Shunt A resistor value with second decimal places (ex. 5.48 ohms. measured around 5 ohms).

d. Connect the multi-meter between Ground (black pin) and Shunt B (blue pin) on the external tool.

e. Record the Shunt B resistor value with second decimal places (ex. 0.53 ohms. measured around 0 ohm).

f. Verify that the value of shunt resistor is the following:

Shunt A - Shunt B = 5Ω ± 2%

10. Connect the multi-meter between Shunt A (red pin) and Shunt B (blue pin) on the external tool.
11. Set the multi-meter to the DC voltage measurement mode.
12. Turn ON the 120VAC and HVDC ENABLE on the Service Switch Panel.
13. Clear the E-Stop condition by pressing the DRIVES RESET button on the Service Switch Panel.
14. Wait about 2 minutes for the gantry display to reset. No flashing lights on the gantry display should indicate this.
15. On the Service Desktop, select [DIAGNOSTICS], then [KV & MA (X-RAY)].
16. Verify the following scan parameters:
    - X-Ray test type = manual
    - Gantry = Disabled
    - Exposure time = 2s
    - No of scans = 1
    - Focal Spot = Large
    - kV = 120kV
    - mA = 50mA
17. Press RUN and take scan.
18. Record the ending voltage on meter to second decimal place(ex. 251.08mV). If this accuracy cannot be obtained, change range on meter and redo scan.
19. Record Console displayed mA value.
20. Repeat the measurement for 200mA.
21. The difference between mA measured (= [measured voltage (mV)] / [shunt resistor value]) and the displayed mA on console should be within ±2% of the requested mA.
22. Turn off all three service switches on the Service Switch Panel (AXIAL DRIVE ENABLE, HVDC ENABLE, 120VAC).
23. Remove the 'JEDI Shunt Measurement Tool' from the generator.
24. Verify mA test point accuracy :
    a. Set the multi-meter to the DC voltage measurement mode.
    b. Connect red signal lead from meter to MA_N (TP20) and black common lead to MA_GND (TP19) on the CT Interface board.
    c. Turn ON the 120VAC and HVDC ENABLE on the Service Switch Panel.
    d. Clear the E-Stop condition by pressing the DRIVES RESET button on the Service Switch Panel.
    e. Wait about 2 minutes for the gantry display to reset. No flashing lights on the gantry display should indicate this.
    f. Launch [KV & MA (X-RAY)] tool.
    g. Verify the following scan parameters:
        - X-Ray test type = manual
        - Gantry = Disabled
        - Exposure time = 2s
        - No of scans = 1
        - Focal Spot = Large
        - kV = 120kV
        - mA = 50mA
    h. Press RUN and take scan.

i. Record the ending voltage on meter to one decimal place (ex. -503.6mV). If this accuracy cannot be obtained, change range on meter and redo scan.

j. Record Console displayed mA value.

k. Repeat the measurement for 200mA.

l. Multiply the meter reading by minus 100. ([TP measured mA] = [measured voltage (mV)] x (- 100)).

* TP measured mA and displayed mA on console must be within ±4% of the requested mA.

25. Turn off all three service switches on the Service Switch Panel (AXIAL DRIVE ENABLE, HVDC ENABLE, 120VAC).

26. Verify kV test point accuracy.

a. Set the multi-meter to the DC voltage measurement mode.

b. Connect red signal lead from meter to KV_N (TP18) and black common lead to KV_GND (TP17) on the CT Interface board.

c. Turn ON the 120VAC and HVDC ENABLE on the Service Switch Panel.

d. Clear the E-Stop condition by pressing the DRIVES RESET button on the Service Switch Panel.

e. Wait about 2 minutes for the gantry display to reset. No flashing lights on the gantry display should indicate this.

f. Launch [KV & MA (X-RAY)] tool.

g. Verify the following scan parameters:

- X-Ray test type = manual
- Gantry = Disabled
- Exposure time = 2s
- No of scans = 1
- Focal Spot = Large
- kV = 80kV
- mA = 50mA

h. Press RUN and take scan.

i. Record the ending voltage on meter to second decimal place(ex. -3.96V) . If this accuracy cannot be obtained, change range on meter and redo scan.

j. Record Console displayed kV value.

k. Repeat the measurement for 100kV, 120kV, 140kV.

l. Multiply the meter reading by minus 20. ([TP measured kV] = [measured voltage(V)] x (- 20)).

* TP measured kV must be within ±3% of requested kV.

* TP measured kV and Displayed kV on console must be within ±2% of requested kV.

5 Finalization

1. Replace inverter and gantry covers if necessary.
2. All the measurements should be within the limits.
3. Record the results on the HHS data sheet.

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